FIU | FLORIDA INTERNATIONAL UNIVERSITY

Department of Electrical and Computer Engineering

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CyberWaster

The Waste Handlers

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I. SUMMARY AND PROBLEM STATEMENT

Abstract

This project aims to look at how waste collection can be improved by using sensors and automation. Most waste system projects only react to how to solve the problem, instead of using data and trends to make decisions. The current approach is inefficient and labor intensive. We are planning to create a simulated robot environment for testing and use those learnings to create a small-scale prototype that will check the waste level, determine when the bin is full, and navigate to a trash disposal location. This project's primary focus is to create the required technology needed for autonomous navigation and to discover the required sensors for collecting waste level data.

Executive summary

This project approaches waste collection as a logistics and optimization problem by integrating sensing, analytics, and autonomous navigation. The system improves collection efficiency while generating useful insights into waste usage behavior. Designed for scalability, the architecture supports residential, assisted-living, and broader smart infrastructure environments. The system autonomously travels from a designated resting location to a disposal point when waste sensors detect that a bin has reached a predefined fill level. This approach reduces manual effort, improves safety, and demonstrates how automation can enable more efficient and data-informed waste management.

Problem Statement

Current waste collection systems rely on fixed schedules and manual observation rather than real-time data, resulting in inefficient labor usage, increased operational costs, and missed opportunities for optimization. Limited visibility into waste fill levels, weight, and usage patterns prevents waste and recycling services from making informed, data-driven decisions. Routine waste handling presents a physical challenge for individuals with limited mobility, including elderly and physically disabled populations. Tasks such as lifting and transporting waste bins require strength, balance, and repeated effort, which can pose safety risks and reduce independence. As the cost of living continues to rise, inefficiencies in time and labor associated with household waste disposal continue to be a consistent task. These combined challenges highlight the need for a solution that reduces physical effort, improves efficiency, and enables smarter waste management through improved visibility and automation.

II. NEED SPECIFICATIONS

Objectives	Specification	Justification
1, 6	An AprilTag signals the robot where to travel.	It automates waste disposal, which would save a person's time and energy.
2	LiDAR sensors detect any blocking obstacles in real time and turn accordingly.	It prevents any collisions that could occur and allows the robot to safely navigate in an environment.
3, 4	An ultrasonic distance sensor alongside 1-4 load cell sensors would measure the levels of waste and signal the robot when it is full.	It ensures that the robot will have accurate detection to correctly decide when traveling is necessary.
5	The data over time of the waste levels would be analyzed for patterns.	It helps optimize schedules for waste collection and efficiently provides a user's waste patterns for collectors.

Objectives
1. The robot should be able to carry trash to a designated location. 2. The robot should be able to avoid obstacles in its path. 3. The robot should be able to monitor waste levels. 4. The robot should be able to detect if the trash bin is full. 5. The robot should be able to comprehend data of the user's waste usage. 6. The robot should be able to reduce the amount of labor for the user.

A. Assumption

- The QR code sticker is placed correctly at destination points for drop off.
- Elderly and physically disabled users are willing to allow an assistive robotic system to support routine waste handling tasks.
- Users will not be required to directly interact with the system during normal operation.

- The LiDAR sensor will detect appropriate sizes of obstacles.

B. Limitation

- Designed to operate with minimal human feedback
- Operation is limited to indoor environments and outdoor concrete or similar flat surfaces.
- The system is not designed to operate on uneven terrain, stairs, or soft ground surfaces such as grass or gravel.
- Performance may be affected by environmental factors such as lighting conditions, obstacles, or layout complexity.

C. Operating Environment

Environment Details	Reference Images
Residential Single-Family Home (Large Property) The system is intended to operate in a residential environment with a paved driveway, landscaped surroundings, and a clearly defined curbside waste collection point. Navigation paths are primarily flat concrete surfaces with minimal elevation changes. Potential obstacles include parked vehicles, outdoor furniture, and pedestrians. Operation is expected to occur in low to moderate traffic conditions.	
Residential Single-Family Home (Suburban Property) This environment represents a typical suburban household with a short driveway and direct access from the home to the curb. The operating surface consists of smooth concrete with grass or landscaping adjacent to travel paths. The system	

must safely navigate near the home entrance, driveway edges, and curbside collection area while accounting for people, pets, and common household obstacles.	
Indoor Facility / Warehouse-Style Environment This environment represents indoor operational areas such as assisted-living facilities and storage areas. The system operates on polished concrete flooring with defined pathways and frequent human activity. Obstacles may include carts, bins, shelving, and temporary obstructions. Reliable obstacle detection and controlled movement are critical due to shared use of space with staff and occupants.	

D. Intended Users and Intended Uses

1. Intended Users:
    - Elderly people (65+)
    - Adults with physical limitations
    - Secondary users such as caregivers and faculty staff

2. Intended Uses:
    - Operates in indoor warehouses or concrete floors.
    - To collect trash and transport it to a drop off location.
    - Reduce labor associated with waste disposal.

III. NEED ANALYSIS

A. Purpose

The purpose of this project is to address inefficiencies in current waste collection practices by shifting from manual, schedule-based operations to a data-informed, and automated approach. Particular emphasis is placed on supporting populations for whom routine waste handling can be physically challenging, such as elderly and physically disabled individuals. By leveraging sensing, analytics, and autonomous mobility, the project demonstrates how waste monitoring and relocation can be optimized through real-time data and intelligent system behavior.

B. Client Need Statement

The client needs a waste collection system that provides real-time visibility into waste fill levels and usage patterns, reduces dependence on manual observation and fixed schedules, and enables autonomous relocation of waste containers. Smart sensors have helped cities dispatch collection trucks only when needed, reducing unnecessary trips and garbage overflow [1]. The company expressed that it reduced unnecessary collections and garbage overflow. Smart bins can optimize collection schedules by detecting anomalies, such as sudden increases in waste volume [2]. The system should specifically reduce physical effort and safety risks for elderly and physically disabled users while improving overall operational efficiency and decision-making.

C. Data Collection

Needs were identified through:

- A structured interview with the project mentor acting as the client
- A survey of potential end users
- Team brainstorming sessions to uncover additional needs

Mentor-Client Interview	
Questions	Answers
On a scale of 1–10, how physically demanding do you find handling waste?	10
How challenging is this task for you to perform on a regular basis?	Very
How often do you find yourself disposing of waste outside your home?	Once a week
How much assistance would make this task easier for you?	Full autonomy
Would you find analytics about your waste behavior helpful in reducing said waste?	Yes
Is the cost of having waste disposed of something you would like to be reduced?	Yes

Potential End Users Characteristics		
Socioeconomics	Age	Primarily elderly adults (65+) and adults with physical limitations; secondary users include caregivers and facility staff.
	Gender	Any
	Income	Expected users would be middle class or higher. This would give an average wage of $56,000+
	Education	Varied educational backgrounds. Systems must be intuitive and not require technical knowledge.
	Disability	Users may have mobility impairments, reduced strength, balance issues, or limited dexterity; the system should minimize physical exertion and risk.
Behavioral	Hobbies	Low to moderate physical activity hobbies. Social hobbies, such as card games with friends and family.
	Reading Interest	General news, lifestyle content, or light informational material. Preference for simple, clear instructions.
	Waste Handling Difficulty	Waste handling may be physically challenging due to weight, bending, balance, or distance to disposal location; users may avoid timely disposal due to effort or safety concerns.
Psychological	Listless	Many expected users would be low energy people, requiring the need for external products to aid in tasks.
	Technology Comfort Level	Varies from low to moderate; the system should feel supportive rather than complex or intimidating.
	Independence	Strong desire to maintain independence and avoid reliance on caregivers.

Team Brainstorming Notes	
Identified Challenge	**Potential Approach or Consideration**
Difficulty moving heavy waste bins	Reduce or eliminate the need for manual bin transport
Risk of falls during waste disposal	Minimize physical handling and walking distance required
Reliance on staff for waste relocation	Enable automated or assisted waste movement
Manual checking of bin fill levels	Provide automated indication of waste capacity status
Desire for resident independence	Support hands-off operation with minimal user interaction
Traffic and obstacle avoidance	High potential of running into people or objects during transit
High potential for missing trash pickup time	A system that is able to transfer waste at a scheduled time or based on environmental condition.

D. Identified Attributes

Main Objective

- Demonstrate relocation systems using simulation and/or a small-scale physical prototype.
- Implement accurate waste capacity sensing to measure bin fill levels.
- Create a simulated environment to demonstrate autonomous robot behavior, system logic, and data flow.
- Collect data to understand the waste usage patterns
- Stay within the project budget

Secondary Objectives

- Define and document a scalable system architecture suitable for expansion to larger environments.
- Design and implement a small-scale physical prototype to validate real-world feasibility.
- Trigger collection and relocation behavior based on predefined waste thresholds.
- Checking the bin weight to prevent overloading the robot with waste

- Limit the movement of the robot if the load is too heavy

Constraints

- Total prototype system cost shall not exceed $800
- Robot shall safely carry a minimum payload of 5 pounds
- Robot shall autonomously navigate to a designated location 15 feet away with a positional accuracy of ±1 foot
- System shall operate reliably on cement flooring and be able to have obstacle detection
- Power system shall support at least two complete autonomous relocation cycle per charge

E. Grouped Objectives

The remaining objectives were grouped into parent categories to clarify design priorities.

Safety

- The system should operate without posing risk to users
- The system should operate without posing risk to pets or local wildlife
- All electrical and battery parts should be protected to prevent any shocks or fire from happening.

Usability

- The system should be easy to operate and understand
- Optional commands to enhance the user experience and improve the product as the user becomes familiar with it
- The system should provide some feedback to the user about the waste level.

Reliability

- The system should function consistently with minimal intervention
- The system should move smoothly and avoid any obstacles in its path.
- The system should be able to handle different terrains without getting stuck.

F. Resulting Problem Statement

This project aims to design an autonomous waste monitoring and relocation system that addresses inefficiencies in manual waste handling while improving safety, usability, and reliability. The system should reduce physical effort and safety risks for users with limited mobility, support independent operation, and provide consistent, data-informed waste handling behavior in residential and assisted-living environments.

The objectives identified through this need specific and analysis provide the foundation for defining the specifications and requirements to ensure the project solves a real-world problem.

IV. FEASIBILITY AND RISK ANALYSIS

A. Technical Feasibility

CyberWaster relies on established robotics, such as the load cells, LiDAR sensors, or the AprilTags. These components are available and can be obtained locally or through online purchases. The system will be using already established frameworks like ROS 2 and Gazebo for simulation. This project relies on existing technologies, making the overall technical risk moderate but manageable.

Below is an assessment conducted by the team ranking the feasibility of set attributes. 5 is a highly feasible option, and 1 is a low-chance feasibility.

Attribute	Score	Why	Solution
Does the technology exist?	5	Yes.	Using existing components like LiDAR sensors, wheels, Gazebo simulation, etc.
Is it available locally or online?	5	Yes, both.	All components are distributed in stores and online stores like Amazon. Some members possess a 3D printer, and FIU has several as well.
Can it be obtained?	4	Yes, however, we require finance.	Our team set price targets for hardware, funding the project ourselves by using open-source software packages.
Are fundamentally new inventions required?	4	Just the 3D structure of CyberWaster.	Design and print the 3D structure of CyberWaster.
How much Technical Risk is there?	3	Ensuring functionality of the hardware and software can be difficult.	Completely build CyberWaster and test its function in different layouts with the simulation.
Total	21		
Average	4.2		

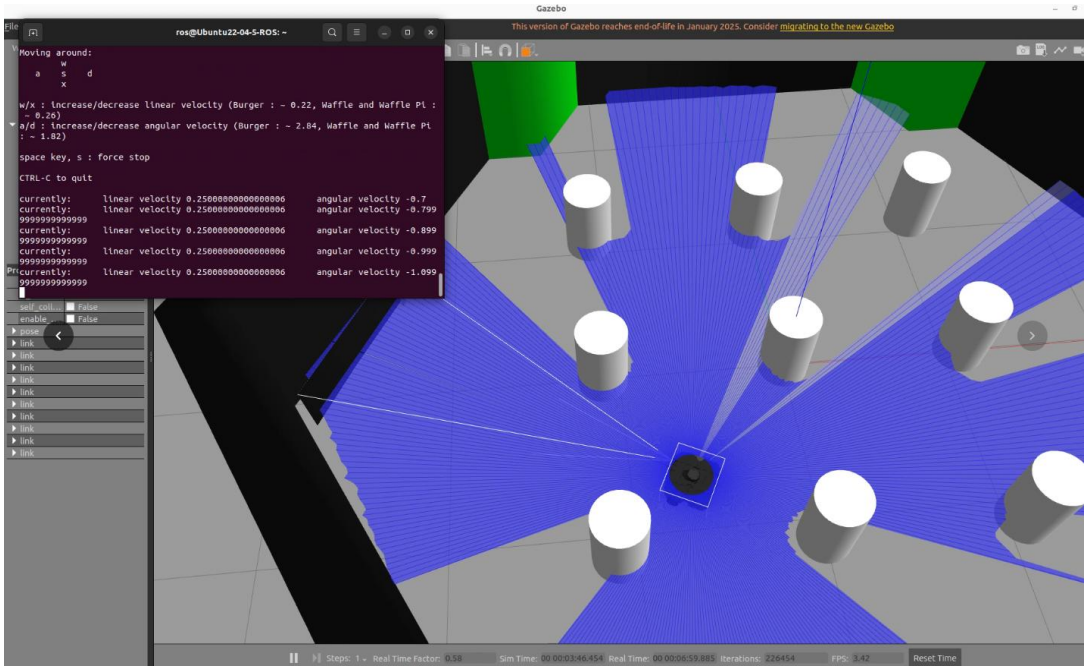

Figure 1: Early Testing with ROS2 + Gazebo during Turtle3 tutorial using lidar on mobile robot

B. Resource Feasibility

Our team has sufficient skills to complete the project; however, there are still some new technologies that are new to us. Some of us have experience in software and hardware development, which assists us in building CyberWaster and creating its simulation in Gazebo. There are still some things we are learning like integrating ROS 2 in this project. Despite the challenges, we are making progress in completing the project, thus making the overall resource risk moderate.

Attribute	Score	Why	Solution
Do we have sufficient skills?	4	ROS2 and Gazebo simulation are new to us.	Our mentor provided some tutorial links, and we are also practicing it as a team.
Do we have sufficient equipment?	5	Yes, we acquired all the components as needed.	Keep components managed during development and testing.
Do we have a sufficient number of people?	5	Yes, 6 members is enough.	The workload will not be too extensive.
How much Resource Risk is there?	4	If any component breaks, it will delay progress due to shipping.	Be careful and treat our hardware components with care.
Total	18		
Average	4.5		

C. Economic Feasibility

This project's system must have a budget to be realistic for a senior project. The required components used in CyberWaster are affordable, such as the wheels, sensors, etc. Additionally, its body is 3D printed, which significantly reduces manufacturing costs. We decided on using simple solutions like using magnets to hold a specific section of the robot together. Since most of the components are not excessively expensive and most of its structure is built through 3D printing, its overall economic risk is low.

Attribute	Score	Why	Solution
Is the project possible, given budget constraints?	5	Most of the components needed, like wheels, sensors, and other electronics, are affordable.	Continue as planned to select cost efficient components.
Do the design choices reduce the overall cost of the project?	5	CyberWaster's body is primarily made of 3D-printed material, significantly reducing manufacturing costs.	Construct the structure from 3D-printed materials to reduce costs.
How much Economic Risk is there?	4	Some components possibly could be damaged during assembly and testing. There is a risk of electric shock.	Handle each component carefully and plan for replacements if needed. Maintain low voltages of less than 24V.
Total	14		
Average	4.7		

D. Schedule Feasibility

We had discussions and decided to assign tasks to each member to keep our pacing organized and efficient for this project. We considered the distinct categories such as the development of Gazebo simulation, hardware integration, software inclusion, and eventually system testing. Even though we will face challenges that may hinder our progress, we believe that fulfilling our assigned tasks will improve our chances of meeting our milestones for the Preliminary Design Review and Critical Design Review. Thus, our overall schedule risk will be considered moderate when considering the complexity of completing CyberWaster.

Attribute	Score	Why	Solution
What are the chances of meeting the immediate mileposts?	4	Our team divided tasks amongst each other to keep organization and progress consistent.	Resume the ordered assignments and track progress with each member.

What are the chances of meeting the PDR (Preliminary Design Review) requirements?	4	Milestones such as the simulation environment and the construction of CyberWaster have been fulfilled.	Keep progressing through the hardware and software integration before the deadline.
What are the chances of meeting the CDR (Critical Design Review) requirements?	3	Some essential features like ROS2 and sensor navigation are still being implemented and may require more testing.	Continue researching ROS2 integration and start testing CyberWaster before the deadline.
Total	11		
Average	3.7		

E. Cultural Feasibility

CyberWaster's purpose is to assist others and make the disposal process more efficient. By reducing the physical effort required to remove waste, our project helps reduce labor and improves the safety of people who struggle with lifting or transporting trash, like elderly people. Assistive technologies have already been implemented to help elderly people in different aspects such as telehealth services, wearable devices, smart home systems, assistive robots, etc. [10] So, the cultural risk is considered low because of how widely assistive technology has become to society.

Attribute	Score	Why	Solution
Will there be a positive impact on the local culture?	5	CyberWaster helps reduce the physical effort that is needed to dispose of waste for those who struggle with it.	Ensure that the design of the system is fulfilled to support accessibility.
Will there be a positive impact on global culture?	5	Assistive technologies are used to support people and improve the quality of life.	Explore methods on how automation helps people with daily tasks.
Will there be potential labor objections to this system?	5	This system intends to help others instead of replacing workers.	Clarify the purpose of assistive technology and this project's goal.
How much Cultural Risk is there?	5	Assistive technologies are accepted worldwide in society.	Ensure that the audience knows how CyberWaster benefits them.
Total	20		
Average	5		

F. Legal Feasibility

This project does not present any significant legal risks as it is being developed as an academic prototype for demonstration purposes. However, when considering our CyberWaster, it will be made of commonly used components that many other service robots use. This ensures that its hardware tools would not just malfunction and explode compared to making a component from scratch. Additionally, our robot is intended to operate in a controlled environment, limiting potential liability concerns. It has been researched that service robots used in social environments may raise legal frameworks like safety; however, existing frameworks can address these issues as technology develops more. [11] Overall, the legal risk is determined to be low due to following basic safety considerations and using standard technology.

Attribute	Score	Why	Solution
Are there organizational conflicts or policies that could affect this project?	5	This project is an academic prototype for demonstration purposes and follows FIU guidelines.	Ensure that the design of the system is fulfilled to support accessibility.
Are there any laws or regulations that could impede the project?	5	All components are the same components that are available in other service robots and do not consist of restricted technologies.	Explore methods on how automation helps people with daily tasks.
Are there any laws or regulations that could limit the project?	4	CyberWaster is meant to operate in controlled environments, reducing limitations.	Clarify the purpose of assistive technology and this project's goal.
How much Legal Risk is there?	5	Our project follows the same current legal frameworks of service robots that focus on safety and liability.	Ensure that all safety standards are followed and do responsible testing practices.
Total	19		
Average	4.8		

G. Marketing Feasibility

Our mobile robot addresses the practical problem of waste disposal for people who struggle doing the labor and elderly people. It can be physically demanding to lift trash and put it away. Since assistive technology is more common in current times, having the ability to have a mobile robot to handle your waste seems plausible for the audience. This project has the potential to be applicable in residential homes or facilities that may need it. The overall marketing risk is low because of the solution to help people in trash disposal and our world's growing trend in

technology.

Attribute	Score	Why	Solution
Will the target audience benefit from CyberWaster?	5	The robot aids people who have difficulty disposing trash, especially elderly and disabled persons.	Complete the system's design to support accessibility and ease of use.
Is there a necessity for a system that assists with waste disposal?	4	Disposing waste can be physically demanding, and service robots can help people perform their daily tasks.	Keep developing the system to solve challenges in waste disposal.
How much Marketing Risk is there?	4	Our system solves a real-world problem and aligns in the growing trend of assistive technology through automation.	Demonstrate the benefits of our project through testing and its final showcase.
Total	13		
Average	4.3		

H. Risk Assessment

A risk analysis is essential for identifying potential challenges that could affect our project of CyberWaster's success. By evaluating risks early, our team can take steps to manage and reduce their impact.

- ★ Technical
  - ♦ T.1: Difficulties with ROS2 integration between hardware and software.
  - ♦ T.2: Inaccurate sensors when detecting obstacles or level of waste.
  - ♦ T.3: Navigation issues when moving to its destination point.
  - ♦ T.4: Motor and Drive Calculations and selection to ft requirements of system
- ★ Resource
  - ♦ R.1: The team's learning curve with ROS2, Gazebo simulation, and robotics.
  - ♦ R.2: Delays if a hardware component breaks during development.
- ★ Economic
  - ♦ E.1: Unfortunate extra costs if components become damaged during development.
- ★ Schedule
  - ♦ S.1: Delays during ROS2 integration with CyberWaster and Gazebo simulation.
  - ♦ S.2: Time conflicts between project work and other responsibilities.
- ★ Cultural
  - ♦ C.1: Users could be hesitant to entrust a robot with their daily tasks.
- ★ Legal
  - ♦ L.1: Possible safety concerns if the robot were to malfunction during operation.

- ★ Marketing
  - ♦ M.1 Skepticism from users about the system's benefits.

		Likelihood of Occurrence		
		Very Likely	Possible	Unlikely
Undesirable Outcomes	Class IV			
	Class III	T1, S1	T2, T3	
	Class II		R1, R2, S2, C1	E1, L1, M1
	Class I			

Risks	T1, T2, T3	R1	R2, E1	S1, S2	C1, M1	L1
Actions	Continue with the project's testing and development.	Learn and practice ROS2 and Gazebo.	Handle all components with care.	Assign tasks and ensure time management is good.	Showcase the benefits of the system.	Ensure operations are safe.

Figure 2: During Testing Confirming Wheel Motion with excess load applied via scale

V. Patents & Intellectual Property

A patent, by definition [22], allows inventors to protect their inventions. They give an inventor the right to exclude others from making, using, selling, or offering the invention for sale, and from importing the invention into the United States for a limited time, and to others if they decide to. It is worth mentioning Patent rights, which are claimed by patents that protect the invention as claimed.

Copyright is the product of the media itself. So, it protects original works of authorship, such as books, music, and films, giving creators exclusive rights to reproduce and distribute their work. Copyright is usually secure from the moment it is created. However, registration is not a condition of copyright protection.

Trademarks are any word, phrase, symbol, design, or combination used to identify the source of goods or services and distinguish them from other sources or products like it. They help people identify the said machine or device in question. While secure and mostly the same, except that it identifies and distinguishes the source of a service rather than a product in question.

There are also trade secrets, which are parts that contain confidential information and features that set them apart from similar products on the market.

Now I'm sure you're asking two questions. How does one go about making a patent, and why is it so important to our project? Well, a patent, like all things, has a process. First, we need to describe what makes our creation unique. What it innovates and so forth. Secondly, the United States Patent and Trademark Office (USPTO) require one to disclose any publication, patent, or other written document that you know to be relevant to your patent. Thus, it is important to decide on the kind of protection of your creation you want (parts, specifics, etc.), which is the third step, and how flexible or inflexible the rules of what you lay down as your copyrighted material can be improved on a later date, or if it's done now, strictly. After which, all you have to do is submit it to the (USPTO) United States Patent and Trademark Office and wait. If it gets approved, you're all done. [23]

As for why it's so important to cyberwaster, having a patent allows it to be global, preventing it from being stolen and sold without permission. It gives us the ability to go 'global' with our robot. Not having a patent means anyone can manufacture and sell the product without giving credit or permission. [24]

As for specifics on which patents are suitable for robots, we found and provided two examples. The first one has basically its assembly copyrighted. Along with specifics on the said parts. Being a robotic arm that must be done in a certain way, even down to its color, since that was its finalized and global design. It does have some flexibility [25]. The second example has a magnetic material itself, along with the circuit board in question, protected by a patent. In fact, most of it is in such detail that one can't even replicate or rip it off in the slightest, which is how a patent protects the invention. [26]

VI. Theory Model Analysis of Potential Ethical Dilemmas of the Project

Theory Model analysis is a step-by-step process of following a particular framework to check on the ethicality of a project. This paper will be using the PLUS framework to break down the CyberWaster project into manageable pieces that will cover any problems that may arise from this project. PLUS is an acronym that means:

- ***Policies.***
    - Does this project follow consistent policies and procedures?

Our project needs to have the creation of policies during development as our 'company' that is running the project did not exist before the development of this project. As such, our policies are liable to change:

- ◊ We are obliged to protect animals and humans, and to help the users of our product for the sake of betterment in their lives.
- ◊ The user will remain in control of the CyberWaster's functionality, serving a supportive role.
- ◊ The decision-making process the CyberWaster makes will be made clear to users to ensure understanding.

- ***Legal.***
    - Is it acceptable under applicable laws and regulations?

The Cyberwaster follows known laws and regulations of Miami-Dade County.
This includes but is not limited to:

1. Leak-proof Design
   The CyberWaster uses a solid design which will not have any leaks during regular use. Our trash can be replaced with little effort or cost since our system is modular. The ability to get a new can and attach it to our CyberWaster will be simple.
   The reason this replacement may be needed is in the case of the trashcan breaking or falling into disrepair.

2. Weight Limit
   The weight of waste in combination with the bin must be less than 50 pounds Current plans for the CyberWaster are limited to a weight of 30 pounds, with a capacity to hold 15 pounds. Current development predicts that the weight of the CyberWaster will be significantly below this range, keeping the overall weight well below 50 pounds.
3. Outdoor Environment
   The CyberWaster is designed to go back inside promptly after it has been emptied, causing it to easily fulfill this law at a higher degree than normal bins used.

- ***Universal.***
    - Does it comply with the universal principles and values of our organization?

It is inherited that the CyberWaster will share the principles of our organization. This is since the organization was formed alongside the CyberWaster. The policies of our organization are formed alongside the CyberWaster. Our top goals are to help humanity by helping those who are not fully able-bodied.

- ***Self.***
    - Does it fulfill our definition of right and good?

Our CyberWaster product follows strict guidelines for the express purpose of helping people. This aligns with our ethical guidelines to help people who are not able-bodied. This product can also help people who are fully able, which gives us an extra incentive to create our product.

VII. Health and Safety

When we are creating a robot, we all think about the health and safety of the users who will be using our robot. A safe robot is designed to not hurt anyone, avoid any obstacle it sees, and have its wire covered so that our user doesn't get burned or shocked. Then there is an unsafe robot where it can harm the users or the environment they are in. For example, it might have exposed wires on them, overheat while working in the sun, or moving unpredictably and break items as it is working. I should mention that our prototype will have its wire showing while we are building and testing it, but the final project will have its wires covered so that the user won't get shocked. Since we are making our robot for elderly people and adults with physical limitations.

Our main goal is to make it easy for our users to avoid getting injured while taking out the trash. We also have a main safety goal which is to reduce the chance of harming our users, their pets, and the environment. Which is why we are planning our robot to slow down or completely stop if it senses an obstacle in its path so that it doesn't hurt anyone. Like for example, they could get back pain from carrying heavy waste or trips on the waste or the bin. We want our robot to help make it easier for our users to handle waste. Our plan with our robot is to help our users by trying to avoid back pains, slips, and even overdoing them. We want to have all the electrical parts for the robot covered because we don't want the user to get shocked while touching the robot. We want to either make labels or cover some parts of the robot so that our users won't get burned to see how hot the material can get while it is in the sun or while it is running.

We are planning to add sensors to our robot so it can detect any obstacles and avoid anything as the robot is taking out the trash. The sensor would help the robot avoid pets, the user itself, and their belongings. Our robot is going to be used outside on a flat surface because if you use it on stairs, grass, or uneven surface the robot wouldn't work properly or damage. This is why we were thinking of creating instructions to have our robot understand the robot limitations and how to use it safely. We also ensure that our robot is made from safe materials, as some people make robots with toxic components that could harm our users. Which is why we want to avoid using that material and ensure users' safety while operating our robots. We hope that our robot will help users handle waste more easily without putting themselves in danger.

As mentioned before, my team wants to ensure our robot is made from safe materials because some of the parts we use can be harmful to our users. We found a research paper called "Ethical considerations in the use of social robots for supporting mental health and wellbeing in older adults in long-term care." [3] This research paper talks about how some robots are made from toxic materials that can be dangerous if anyone touches or inhales it over a long time. This made us realize we need to carefully choose the materials we use, so we don't expose our users to illness or health problems later. Since we plan to build our robot using 3D printing, we thought it best to research the associated risks. We found an article called "The

Health Effects of 3D Printing" [5] that explains that the common 3D printing material, ABS, releases tiny, harmful chemicals that, when heated, can irritate our lungs and cause health problems later. While PLA is the safer option to use because it produces fewer chemicals into the air. Based on two sources, we need to choose carefully on which material we are going to use because we want our robot to be safe to use and not harm anyone who uses it.

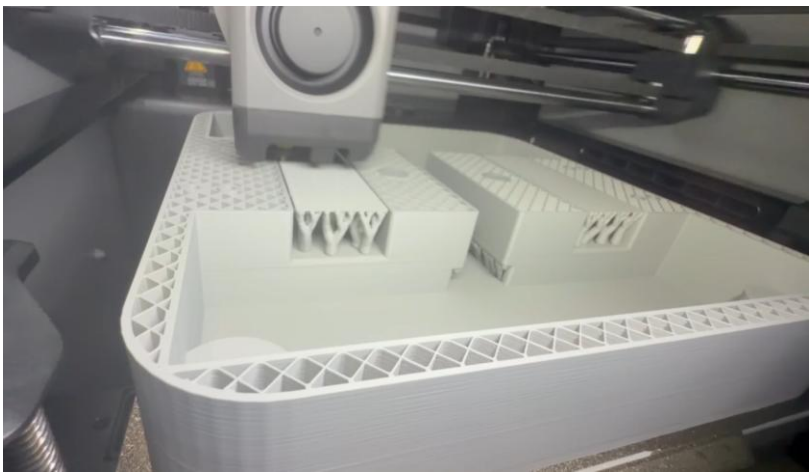

Figure 3: Photo take during 3D printing of our prototype CyberWaster

Materials	Safe to Use	Risk if Uses
Plastic	Need to choose a non-toxic plastic that doesn't release chemicals when heated.	All plastics will release harmful chemicals when its heated it up that will cause irritation on the lunges
3D Printing	Maybe PLA because it produces less harmful chemicals in the air.	ABS produces a lot of styrene and tiny particles that will irritate your lungs and cause health problems.

As we discussed before, we don't want our robot to hurt anyone, which is why we use sensors to avoid obstacles and cover its wires, so users won't get shocked or burned. We found another research paper called "Robotics to Enable Older Adults to Remain Living at Home", which talks about how robots are helping older adults to stay safe at home by helping them avoid falling and moving around easily. [4] It does mention how it helps monitor its environment and reminds people to take care of themselves, like a service dog. This paper helps us better understand how we want to make our robot, because we need it to notice its surroundings so it can know where the trash is and whether there are any pets around. Also, it helps us understand more on how to make our user safe as we are making the robot. To show how our robot is making the decision while moving, we created a flowchart below.

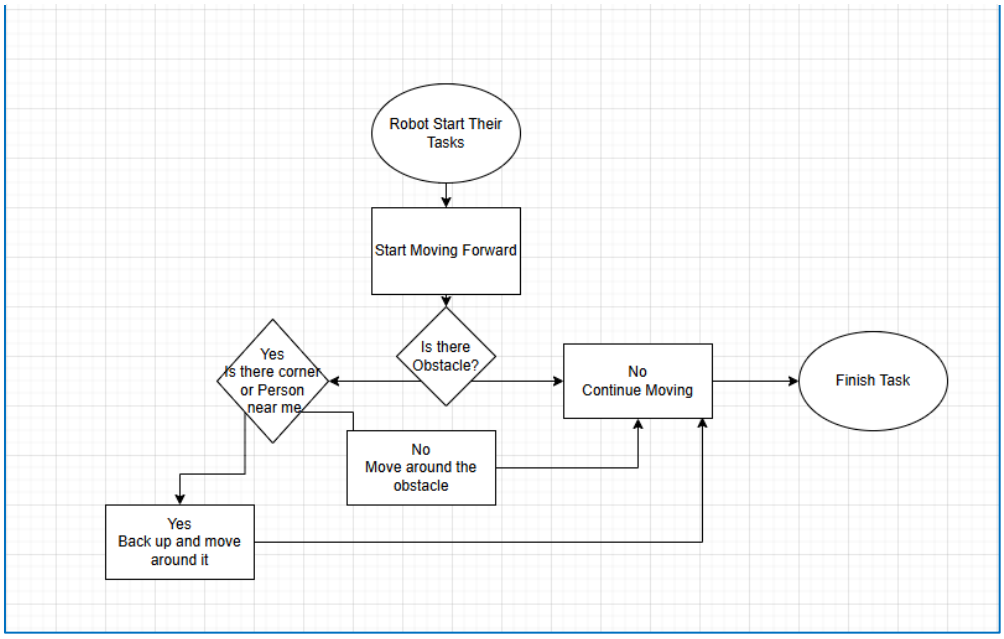

Figure 4: This flowchart shows how the robot is deciding what to do when it senses something in its path.

IP (Ingress Protection) [21] ratings are a standardized classification system used to define the level of protection an enclosure provides against the intrusion of solid particles (such as dust) and liquids (such as water). This is highly relevant to the CyberWaster system, as the device may operate both indoors and outdoors, exposing it to environmental contaminants. During operation, particularly when waste is transported, dust and debris may be disturbed, requiring adequate sealing to protect internal electronics and mechanical components. Additionally, operation in potentially damp or wet environments introduces risk to electrical systems, making moisture protection critical.

From a design and compliance perspective, selecting an appropriate IP rating supports reliability, safety, and regulatory alignment, including potential insurance and consumer safety requirements. An IP67 rating is a strong candidate for this application, as it ensures complete protection against dust ingress and provides resistance to temporary immersion in water. This level of protection enables the system to function safely in outdoor conditions such as rain exposure while maintaining the integrity of internal components.

Overall, we want our robot to keep the users safe and healthy while taking out the trash easier for them. We just want to make sure that the wires are covered; we are using non-toxic materials that won't affect the elderly and people with physical limitations later in life and prevent any shock or burn when touching our robot. Also, we need to make sure that our robot knows its surroundings before moving around because we don't want the robot to hit anything or anyone as it is running. Our focus on our project is to have our robot help with waste without causing any issues to our users and feel safe around it.

VIII. Sustainability and Standards

Compliance with established robotics and machinery safety standards is a critical aspect of the system design, particularly due to the presence of autonomous motion and operation in environments where humans or obstacles may be present. One of the primary standards considered is ISO 10218 [7] which defines safety requirements for industrial robots and robotic systems. While the system is not a traditional fixed industrial manipulator, meaning a stationary robot arm mounted in one location, the core safety principles outlined in this standard are still applicable due to the use of motors, sensors, and autonomous control. We believe these principles remain highly relevant, especially since the system presents similar risks related to motion, actuation, and interaction with its environment. These ideas also align with broader research in mobile robotics where safe navigation and predictable system behavior are necessary for operation in real-world environments.

A key requirement defined in ISO 10218 [7] is risk assessment and hazard identification, in order to align with this, the system could be designed with consideration of potential failure conditions such as unintended motor activation, communication failure between control components, sensor inaccuracies, and possible collisions. These risks can be addressed through the use of the Robot Operating System (ROS), which allows for modular communication between system components and supports continuous monitoring of sensor inputs and control signals. For example, sensor data from LiDAR and ultrasonic modules could be continuously verified, and any weird readings might trigger a system response such as stopping or adjusting movement. We think incorporating this type of monitoring is important for maintaining safe operation, particularly in environments that may be unpredictable.

Another important requirement outlined in ISO 10218 [7] is safe motion control. The standard in this case emphasizes limiting speed, acceleration, and overall system behavior to reduce the likelihood of injury or damage. To reflect this, the drive system would incorporate defined limits on motor output with motors being controlled using PWM and potentially feedback-based control to maintain predictable movement. Within ROS, motion commands are regulated through global and local planners. The local planner continuously generates collision-free trajectories and adjusts velocity commands using real-time feedback, resulting in smooth and stable motion rather than abrupt transitions. This is especially important for a mobile platform operating near obstacles or people. [29] These algorithms often come in a format that will append extra path onto the given path instead of needing to start a new path or recalculate the path for any found obstacles [27].

Continuing on, the standard also highlights the importance of protective measures and safeguarding. In fixed industrial systems, this is often achieved through physical barriers, but for a mobile platform, safety relies more on sensor-based detection. LiDAR would be used in our case as the primary sensing method to continuously scan the environment and detect obstacles in real time. This data can then be processed within ROS to generate maps and determine safe navigation paths. Ultrasonic sensors could also serve as an additional layer of detection for close-range obstacles, providing redundancy in case of LiDAR limitations.

Emergency stop functionality is another requirement emphasized in ISO 10218 [7] so to address this, both hardware and software-based stop mechanisms could be incorporated. A physical kill

switch could disconnect power to the motors while a software-based stop command within ROS might override active processes and bring the system to a controlled stop.

In addition to ISO 10218, ISO 12100 [8] provides general guidelines for machine safety, particularly in the areas of risk assessment and risk reduction. This standard supports the idea that hazards should be minimized at the design stage rather than addressed later. For example, limiting motor torque and maximum speed could reduce the severity of potential impacts, while thoughtful placement of components may help reduce mechanical or electrical risks.

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